Useful information:

- for jailbreaks / illicit behavior, people have tried training models (adversarial training) and other variants, but in the industry, openai and Anthropic have seen the most success by just training really good specialized classifiers for detecting harm
- Similarly, we might see more success

Constitutional Classifiers for Values	
Primary research question (One question with an answer you could recognise, such as "Does X change Y in setting Z?" or "Which of three options is most feasible, and why?")	**Workshopped Question:** Can we adapt *'Simko et al.'s'* (6.11.2026) new constitutional classifier architecture to readily adapt to variable value categories? **Sub-question(s):** Can we adapt *'Simko et al.'s'* (6.11.2026) new constitutional classifier architecture to any of the value categories* by finding and training a classifier on a suitable dataset? Does our result satisfy the desiderata within Anthropic's constitutional classifiers: Desiderata 1: Robustness to universal jailbreaks Desiderata 2: Practical deployment viability Desiderata 3: Flexibility (*safeguards should be flexible enough to adapt to evolving domains of potential harm)* For the categories above (we will try multiple categories), does our constitutional classifier successfully detect behaviour that is unaligned with human values, even on out of distribution samples? *For example, immoral behavior, biases (gender bias, etc), historical revisionism, "non-aligned" moral judgement, authoritarian values endorsement, human rights, and also potentially alignment to European values (e.g, detecting pro-russian speech and outputting a pro-Ukraine view), reward hacking etc.

Context and sources (What is already known, in your own words, with five to eight cited sources as links. End with what already exists and where your project adds something)	Existing constitutional classifiers focus on harms such as CBRN weapons, violence and illegal activity. Value-related categories are more context dependent (a sentence may or not be deemed acceptable depending on the situation), and there is little known work on whether a constitutional classifier works on such categories. Our work hopes to produce a value-related classifier. We intend to share everything needed to reproduce our results and extend them. Related work on classifiers: • Constitutional Classifiers: Defending against Universal Jailbreaks • Constitutional Classifiers++: Efficient Production-Grade Defenses against Universal Jailbreaks • EuroSafeAI — safe.eu/index Papers from PhD supervisor Zhijing: • Democratic or Authoritarian? • When Do Language Models Endorse Limitations on Human Rights Principles? • Preserving Historical Truth: Detecting Historical Revisionism in LLMs • Language Model Alignment in Multilingual Trolley Problems • Are Language Models Consequentialist or Deontological Moral Reasoners? • When to Make Exceptions: Exploring Language Models as Accounts of Human Moral Judgment • Simulating Democratic Deliberation: Representing Pluralistic Preferences through Electoral Systems in Multi-Agent LLMs • Evaluating Second-Order Bias of LLMs Through Epistemic Entitlement • How Does Alignment Tuning Shape Representations of Sycophancy and Related Cue-Induced Biases in LLMs? • https://proceedings.neurips.cc/paper_files/paper/2022/hash/b654d6150630a5ba5df7a55621390daf-Abstract-Conference.html • Link to literature folder in GitHub repo: https://github.com/Sid-Diamond/Proj-IFR/tree/main/literature
Core deliverable (What is the minimal-work version of your project that would provide some value?)	**1. Constitution:** a document defining what violates the category and what is explicitly allowed • We have an existing constitution(al classifier architecture) released on 6.11.2026. Information will be sent out to us. • We will adapt it from the accompanying repo released alongside the paper.

	2. Dataset: prompt-response pairs generated from the constitution that is checked by an LLM judge. - 'generated' (probably not what we will do!): 1. Use *'helpful only'* model to generate queries that span the constitution categories. 2. Use '*helpful only*' model to create answers, both harmful and harmless, for each synthetically generated input. 3. Use *'prompted-llm*' to filter out any inadvertent refusals from the generated input-output pairs. - Benchmarks exist but we can either use them for training or testing. - We should look for a dataset for a set of categories, if we find cool ones then we can adapt our requirements. - I think it is more likely we will find a dataset on hugging face that has a good licence and we can argue that a strong training vs eval split is tractable. - This will probably be the longest, and most boring part of the project!! **3. Classifier:** a small open-weight model fine-tuned with LoRa as (an exchange AND output / input) classifier ● Use linear probes (https://arxiv.org/pdf/1610.01644) in combination to escalate harmful prompts to the classifier: 1. Pass labelled data through llm. 2. Examine activations at each layer and label these with the same label as the data passing through llm. 3. Per layer, fine-tune a logistic regression model to predict the probability of the label; i.e. P(y_i=1 | activation vector h) = w_T*h_i +b. i.e. use the data to fit the parameters of the weight vector w and the bias term, b, given an activation vector h_i 4. Put this probe (regression model for a given layer) on a 'slider', so when an input is given to the llm, it "slides" to the most "activated" layer for harm. If here P(y_i=1 | activation vector h) > C. Where C is some cutoff value. **4. Evaluation:** evaluate performance of classifier on test dataset. ● Presumably something simple like an F1 on harmful data? ● Application of known red-team tests? ● Best of N attack ?

	5. Output: codebase, dataset, and report.
Target and stretch (What you expect to deliver by week eight, and what you would add with more time)	**Target (week 8):** core deliverables **Stretch Goal:** An add-on to our Open-Source constitutional classifier work (EuroSafeGuard) that protects and defends against other types of immoral behavior, anti-democratic values, aligns model behavior to human rights, on top of our initial system that mostly focuses on illicit behavior (CBRN, self-harm, etc). The classifier should detect when the input / output should be intervened on, and steer it accordingly with re-prompting with an additional flag. By week 8: an initial implementation of the system. All data in the system. Good preliminary performance on benchmarks such as MoralExceptQA, human rights, gender bias.
Riskiest assumption (What could sink the project, how you will test it by the end of week two, and your fallback if it fails)	**Riskiest assumption:** that the category can be defined clearly with consistent labels, as to what is allowed / violates the value. **How to test this:** using LLM as a judge or manual verification **What is the fallback:** existing datasets already label the samples so we can use those.
Roles (For each resident: their role, the skills they bring, one stretch goal, and weekly hours - at least five each)	**Katie:** - **role:** ML researcher - **skills**: has experience with jailbreak research and model training - **stretch goal**: contribute more than 10 hours - **weekly hours**: 5-10 a week **Sid:** - role: ML Researcher - skills: Research experience in an academic environment, fine-tuning, dataset evaluation analysis - stretch goal: Produce STATO work at a professional standard - weekly hours: 12-15 a week **Muhammed:** - role: ML Researcher - skills: Experience communicating AIS research/distilling topics & attention to detail

	- stretch goal: Fully upskill & contribute some key insights (from my current starting position) - weekly hours: Uncapped (10–20/week)
Impact pathway (What is the project's Theory of Change? How would finding an answer to this question could be valuable/ impactful)	Currently, SOTA AI systems are protected with constitutional classifiers (Anthropic, OpenAI). These systems are very safe and hard to bypass (FAR.AI, 2026) but mostly focus on illicit behavior. While the OpenAI constitution does mention topics light human rights etc, most evaluation is on CBRN. We will contribute to the newest open-source constitutional classifier implementation (EuroSafeGuard) by adding. We will release the constitution, data, and evaluation protocols so that others can reuse and extend this research. External classifiers are one of the existing safeguarding methods that can be used to improve model safety.
Growth and extension (Two or three follow-on work packages, each small enough for one person to progress part-time, and organisations that could use or host them)	Post-train open-weight models using our dataset to produce more aligned models. Use Multi-agent debate for certain morality topics. This can be used as a stronger judge than a single model.

Project Plan Template

1. Read through all the recommended papers
    a. set up the repository
2. Draft the constitution
    a. start producing the dataset
    b. use judge to verify dataset
    c. Separate the test split
3. Train the classifier with LoRA
4. Evaluate the classifier
5. Start working on paper
6. Work on extensions?

Week	End of week output	Responsible Resident	Check-in questions?
1 26 Oct - 1 Nov	Project Proposal and repository set up.	All	Is the project proposal submitted in a respectable state?
2 2–8 Nov	Finish the literature review. Find datasets. Get familiar with the classifier that will be released on 6/11/26. Some of this research can be used for our research.	All	Do we have at least two potential datasets? Have we all familiarised ourselves with *'Simko et al.'* 6.11.2026? Have we discussed the size of the dataset, time constraints and licensing constraints that determine the viability of the dataset?
3 9–15 Nov	Edit the 'viable' dataset into something directly usable. Run a simple test on the 'viable' data	All	Are we ready to begin outlining a project report?
4 16–22 Nov	Go, adjust or pivot decision made; report outline agreed Create our architecture adjustment on the 'viable' dataset and work on the core code in the repo Perhaps we split the codebase into auditing for edge cases and preparing the code to be readily generalisable to the	All	Are we at a point in the code where the core deliverables are looking achievable in the time frame? Are we making sure the code is up to standard? In a week's time how

	value categories? Perhaps split the group further and someone continues looking for the next set of data so we can readily adjust?		likely is it we will be able to write an MVP?
5 23–29 Nov	Continue creating our architecture adjustment on the 'viable' dataset and work on the core code in the repo	All	Could we write a report using the results we have thus far? If not, why not?
6 30 Nov–6 Dec	MVP complete and scope frozen; first full draft started	All	Does everybody feel happy with the section they're writing? Does the scope look appropriate? Do we have a healthy and effective report auditing/ feedback system in process? Are we being realistic with how we've framed the project?
7 7–13 Dec	Full draft with the mentor for feedback	All	Does it look like an extension is materialising? Have we left enough time to discuss this in week 8 or is it all hands on deck hitting the core deliverables?
8 14–20 Dec	Final report, including next steps for extension; plan for sharing	All	